Fiscal Year:	FY 2017	**Task Last Updated:**	FY 05/18/2017
PI Name:	Stanton, Patric Ph.D.		
Project Title:	Effects of Space Radiation on Long-Term Synaptic Plasticity and Neurogenesis in Normal and Alzheimer's Disease Transgenic Mice		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	patric_stanton@nymc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	914-594-4883
Organization Name:	New York Medical College		
PI Address 1:	Cell Biology and Anatomy		
PI Address 2:	Basic Sciences Building, Room 217		
PI Web Page:			
City:	Valhalla	**State:**	NY
Zip Code:	10595-1690	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	11/01/2012	**End Date:**	10/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2017 per S. Monk/LaRC (Ed., 9/20/16) NOTE: Extended to 10/31/2016 per S. Monk/LaRC (Ed., 9/14/15)		
Key Personnel Changes/Previous PI:	October 2016 report: No Changes.		
COI Name (Institution):	Blakely, Eleanor Ph.D. (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	NNX13AB66G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The hippocampus and dentate gyrus are critically important brain regions necessary for the formation of long-term memories, and damage to these regions is a cause of memory deficits in early Alzheimer's disease patients. Both regions are also notable as sites where stem cells can differentiate into new neurons throughout life, a process called 'neurogenesis.' An abundance of data indicate that exposure to even low doses of ionizing radiation prevalent in space travel (SR) results in impairments in learning and long-term reduction in hippocampal neurogenesis. It is unknown how radiation causes these impairments, whether they predispose individuals to or accelerate the development of Alzheimer's disease, and how we can prevent such sequelae. The key, central objectives of this proposal are to 1) examine the long-term impact of SR on electrical communication between neurons, and changes in function and fine structure of the sites known as 'dendritic spines' where neurons form synaptic contacts to communicate with one another; 2) determine whether SR, in reducing neurogenesis, also alters the functionality of newly-born neurons; and 3) determine whether, in the most widely-studied animal model of human Alzheimer's disease, SR differentially affects electrical or physical function of neurons, and/or accelerates the disease process. The methods we propose to accomplish these goals are 1) electrical recordings from living slices of mouse brain, and optical imaging of minute compartments in neurons known as dendritic spines; 2) in vivo injection of fluorescent markers of newly-born neurons that arise from the continual process of neurogenesis; and 3) comparison by the same methods of the effects of SR in slices from normal wildtype mice to slices from the most widely-used transgenic mouse model of human Alzheimer's disease. Our proposed studies directly address multiple key objectives in this RFP (request for proposals), namely to determine dependencies of late central nervous system (CNS) risks such as early onset of dementia or Alzheimer's disease, determine if SR causes early molecular, cellular, and tissue environment changes in hippocampus indicative of dementia, and to develop new biological models of Alzheimer's disease or other late forms of dementia that occur in humans. The existing knowledge gap is immense and presents a major obstacle to rational assessment of short- and long-term risk to the central nervous system of low-dose radiation exposure in space. Our experiments will, for the first time, examine the mechanisms by which SR impairs synaptic function in normal brain, assess whether SR does, in fact, enhance long-term risk of Alzheimer's disease, and fuel the search for countermeasures to reduce this risk, towards making the environment of space safe for long-term exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal addresses key questions of susceptibility of normal and diseased brain to subtle yet critical long-term damage from radiation. For humans to spend long periods in space, we must clearly understand the CNS risks of space radiation to brain function, using direct measures of synaptic function, plasticity underlying cognitive learning and neuronal replacement, and identify effective countermeasures. However, the brain is very susceptible to radiation, including medical and occupational exposure on Earth. There are pluripotent stem cells in the CNS, and key brain regions experience replacement of neurons by stem cells throughout life. Since stem cells are very sensitive to radiation, even low dose radiation could cause subtle alterations in CNS function, and simultaneously impair its ability to repair damage. If radiation accelerates the development of neurodegenerative diseases such as Alzheimers disease and alters brain cognitive performance during a mission, effective countermeasures will be critical to mission success. Transgenic mouse models of Alzheimers disease develop the disease in months (far more rapidly than humans), enabling us to determine whether radiation exposure accelerates disease progression, and begin testing antioxidants as potential protectants.
Task Progress:	No final report received. For related papers published after the project ended, please see the Cumulative Bibliography (Ed., 5/6/26). Under our previously funded project, we engaged in four irradiation campaigns at Brookhaven National Laboratory with beamtime granted for over 100 mice per campaign. We have accrued large enough sample sizes to study the variables proposed in our specific aims, namely: ion species-specific, age-specific, sex-specific, time-specific, and strain-specific effects of galactic cosmic radiation (GCR) on hippocampal neurogenesis, synaptic plasticity, and learning and memory. In the last year, under an unfunded extension of our project, several of our cohorts aged to their assigned timepoints, and we have made significant progress in fulfilling many of our proposed experiments. Below is a summary of new results which, together with previously acquired data, are being prepared for submission for publication. Characterization of long-term effects of 56Fe exposure on neurogenesis, long-term potentiation of synaptic strength in hippocampal slices, and learning and memory: Antibody labeling of newly born neurons in hippocampal sections from male and female mice exposed to 56Fe at a dose of 10, 50, or 100 cGy was used to quantify neurogenesis 2 months post exposure and at longer timepoints (6, 12, 20 months) post exposure. We also studied activity-dependent changes in neurophysiology in hippocampal slices in control and irradiated mice at the same timepoints. Spatial learning, memory, and cognitive flexibility were assayed using the active place avoidance task. In this task, a progressive decrease in the number of entries into a stationary shock zone on a rotating platform is an indication of avoidance learning. Re-assigning the shock zone to a different position on the platform forces the mouse to discriminate between two similar experiences, eliciting cognitive discrimination of the contexts (re-learning). Characterization of long-term effects of 28Si exposure on long-term potentiation of synaptic strength in hippocampal slices and learning and memory: Similar to the experiments in which we exposed mice to 56Fe, hippocampal slices from male and female mice exposed to 28Si at a dose of 10, 50, or 100 cGy were used to study synaptic transmission and neurophysiology. Spatial learning, memory, and cognitive flexibility were assayed using the Barnes Maze. Latency to find a target hole connected to an escape box during learning trials across at least 4 days in the maze was used as a measure of learning. Memory was tested by measuring the amount of time spent in the target zone on after the escape box was made inaccessible. Establishing a model to study chronic effects of GCR on neurogenesis and functional integration of adult-born neurons in the hippocampal dentate gyrus:

	We have acquired a strain of mouse which, based on the Cre/loxP recombination technique, can be used to pulse-label neural progenitor cells and adult-born neurons. This recombination induces expression of fluorescent YFP (yellow fluorescent protein) selectively in newborn neurons. These neurons can then be quantified to measure the rate of proliferation before or after irradiation, and patch-clamped for whole cell electrophysiological recording.
Bibliography Type:	Description: (Last Updated: 03/10/2021)
Articles in Peer-reviewed Journals	Miry O, Zhang XL, Vose LR, Gopaul KR, Subah G, Moncaster JA, Wojnarowicz MW, Fisher AM, Tagge CA, Goldstein LE, Stanton PK. "Life-long brain compensatory responses to galactic cosmic radiation exposure." Sci Rep. 2021 Feb 22;11(1):4292. https://doi.org/10.1038/s41598-021-83447-y ; PMID: 33619310; PMCID: PMC7900210 , Feb-2021